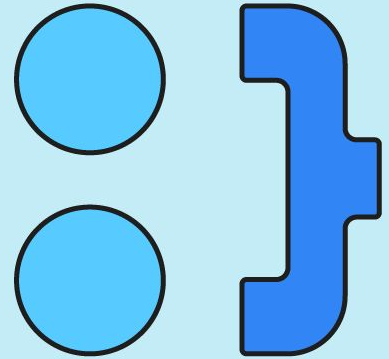
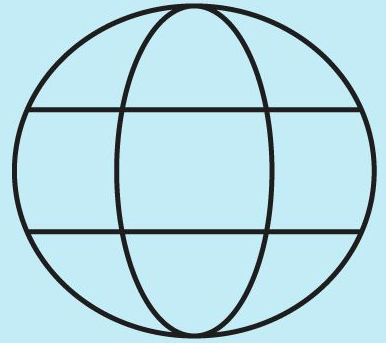




Presentation: Kickstart your Web Journey: Intro to Web Development and HTML



Google Developer Group
Jomo Kenyatta University of Agriculture and
Technology

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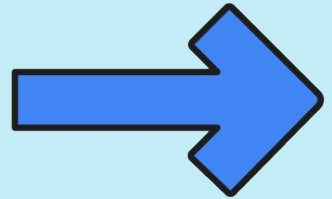
- a) Intro to Web Development
- b) How the Internet Works
- c) Intro to Frontend Development
- d) What is HTML
- e) Creating your first HTML file (Hello World file)
- f) Tags, attributes, case insensitivity, entities, comments and whitespaces





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Intro to Web Development



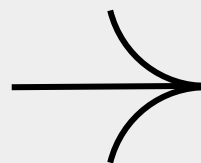
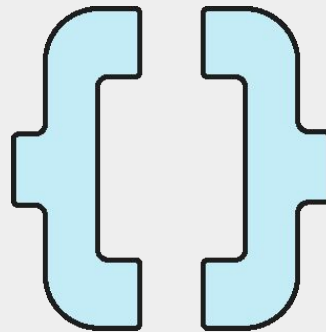
What is Web Development?

Web Development is the process of creating and maintaining websites that run on the **Internet**.

Users access websites through **web browsers**, the browser sends a **request** to the **hosting server**, which processes the request and returns a **response** back to the **client**, using either **HTTP/HTTPS**.

Much of the technical jargon and terminologies used above will be discussed in detail in the next chapter.

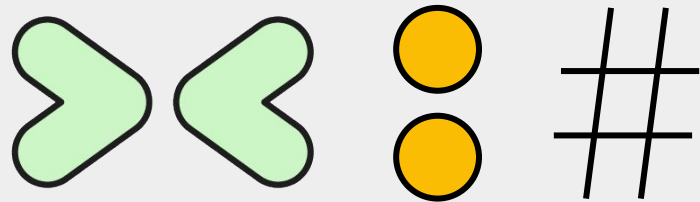
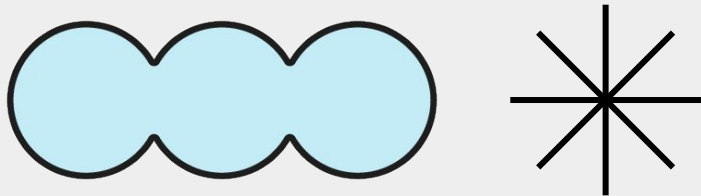
Web Development is split into 2 major areas: **Frontend Development** and **Backend Development**.



What is Frontend and Backend Development?

Frontend Development.

This area covers client-side technologies used to structure, style and make web pages functional and interactive within the user's browser.



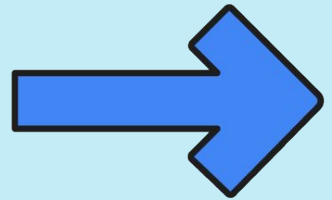
Backend Development.

This section covers server-side technologies that work behind the scene to handle data, run the website and store information.



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How the Internet works.





Video Summary pt 1.

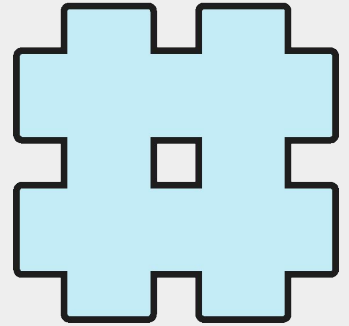
Internet - worldwide interconnected system of computer networks.

Client - computer that requests data from a server through the internet.

Server - computer that hosts, processes and responds to user requests over the internet.

Browser - software application used to access, retrieve and view information on the internet

Hypertext Transfer Protocol - rules used by browsers and servers to communicate and transfer data across the internet. HTTP is unencrypted, while HTTPS is encrypted.



Video Summary pt 2.

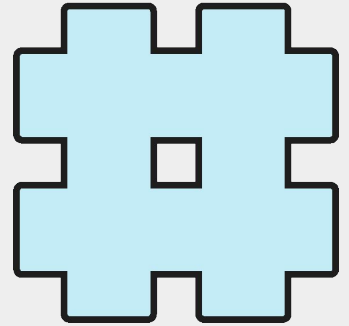
Request - message sent by a client to a server.

Response - message a server sends back to a client after receiving and processing a request.

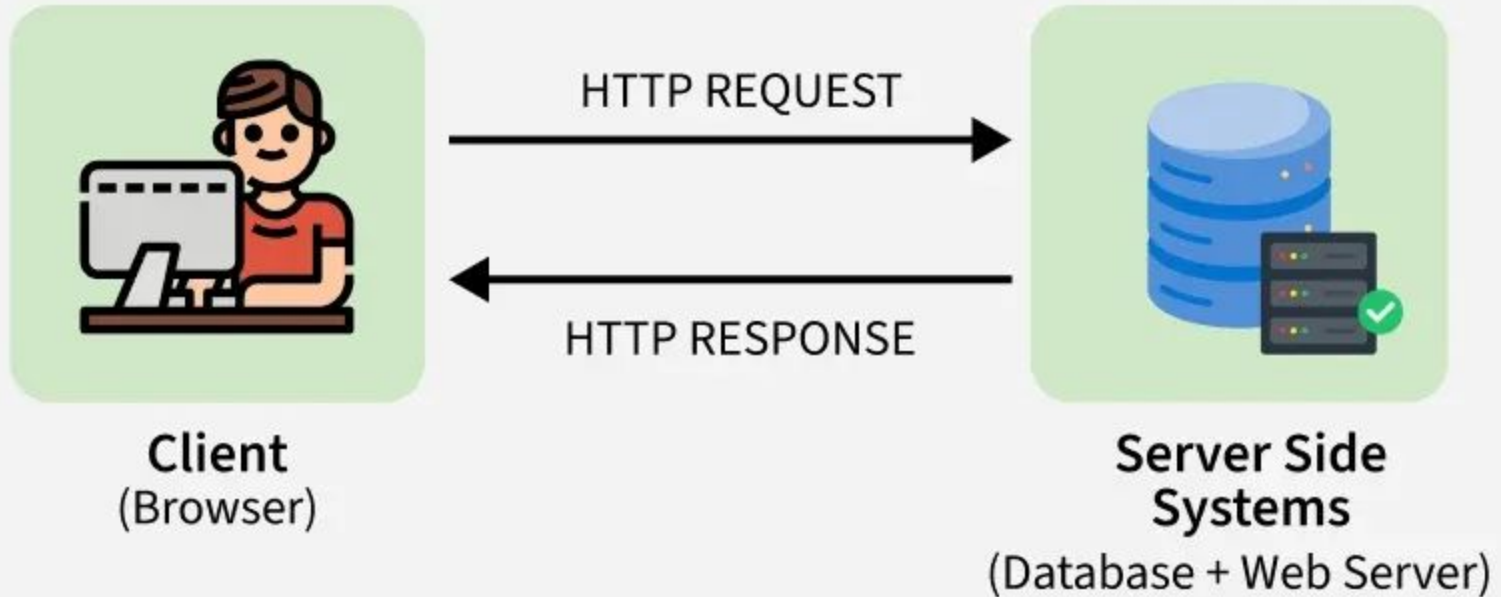
Internet Protocol (IP) address - a unique identifier to every device connected to a network.

Domain name - human-readable web address that users type into a browser to visit a website.

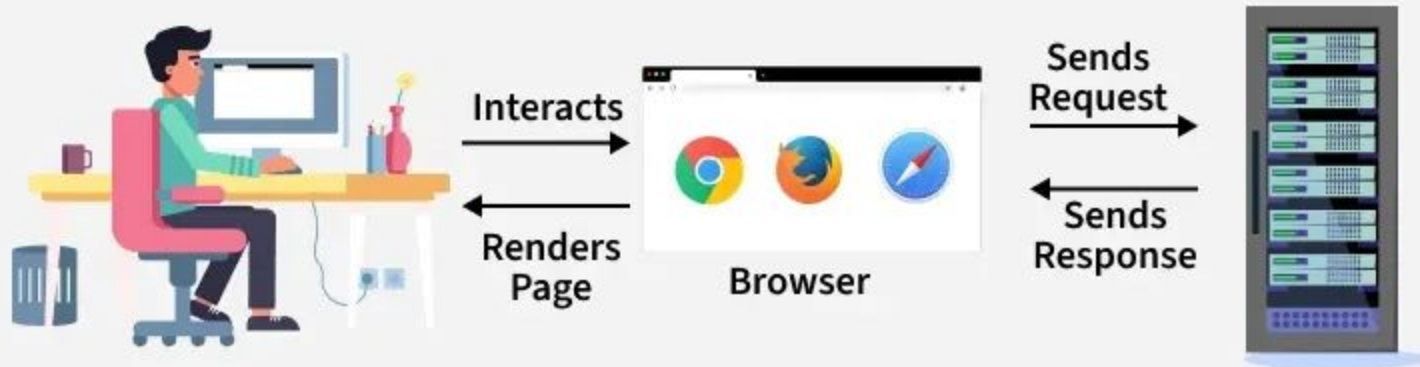
Domain Name System - phonebook of the internet that turns website names into machine readable IP addresses.



Request and Response Cycle



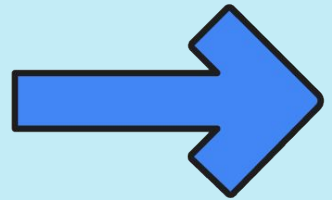
Web Browser





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Intro to Frontend Development.



Frontend Technologies



HyperText Markup Language (HTML)

Language used to create the basic **structure** and content of web pages.

Cascading Style Sheets (CSS)

Language used to style HTML content.

JavaScript (JS)

Language used to add interactivity and functionality to web pages.



HTML Basics

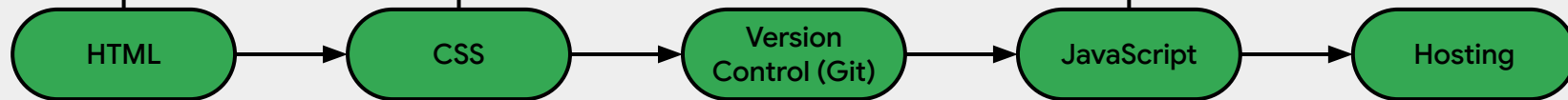
Intro
Tags & Attributes
Media
Forms
Semantics
Accessibility

CSS Frameworks

BootStrap
Tailwind CSS

JS frameworks

React
Next
Angular
Vue
Express



Intro
Box model
Layouts
CSS Display
Responsive design

CSS Basics

Intro
Functions
Data types
Events
DOM

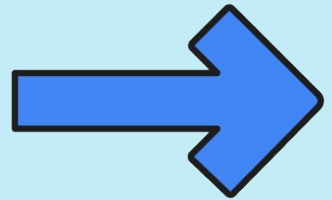
JS Concepts





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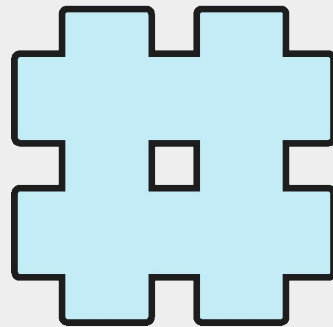
What is **HTML**?



What is HTML?

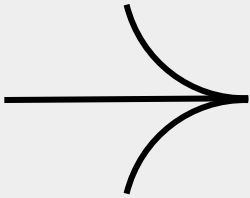
HTML is the standard language used to create and structure the content of web pages. It is not a programming language, but rather is a markup language that used a system of tags to annotate text, images and other media so that browsers know how to display them.

It uses tags, written inside angle brackets < and >, to create elements. Most elements require both an opening and a closing tag to wrap around content.



Fun fact.

It was created in 1993 by Tim Berners-Lee. It originally contained 18 tags. Currently, the modern standard (HTML5) has over 150 tags.



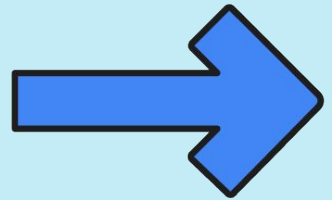
1993
Year of creation

140+
Number of tags in
the modern standard



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Create your first **HTML** file!



Before We Start:

- a) Open VSCode/ your IDE.
- b) Install Live Preview Extension.
- c) Create a folder called Hello World in HTML.
- d) Create a file called index.html.
- e) In your terminal, initialize a git repository (git init), stage the new folder with the file (git add .), and commit (git commit -m "Initial commit"). You can optionally push that code.
- f) Open your index.html file.



index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
    content="width=device-width,
    initial-scale=1.0">
  <title>My First Page</title>
</head>
<body>
  <h1>Hello World!</h1>
</body>
</html>
```

